



Estimating numerical calculations

Task A: Estimating using significant figures

1) Fill in the gaps to make the approximations correct. You can only use each number once.

250 200 4 4 900 50 5

- a) $4.78 \times 927 \approx 5 \times 900$
- b) $3.892 \times 893 \approx 4 \times 900$
- c) $203.38 \div 4.302 \approx 200 \div 4$
- d) $245.29 \div 47.29 \approx 250 \div 50$

2) To **estimate** this,

$$19.173 \times 338$$

a) Which of these calculations would you use? Explain your choice.

20×350 20×300 19×350 19×300

There is no 'right' answer to this question. As long as there is justification to the calculation chosen.

For example, 20×350 is acceptable as 19.173 has been rounded to 1 s.f. and 338 has been rounded to the nearest 50.

20×300 is acceptable as 19.173 has been rounded to 1 s.f. and 338 has been rounded to 1 s.f.

19×350 is acceptable as 19.173 has been rounded to 2 s.f. and 338 has been rounded to the nearest 50.

19×300 is acceptable as 19.173 has been rounded to 2 s.f. and 338 has been rounded to 1 s.f.

b) Which of these gives the closest **estimate**?

*19×350 gives the closest **estimate**, since 19.173 is closer to 19 than 20 and 338 is closer to 350 than 300.*

c) Which gives the **estimate** that is furthest away?

*20×300 gives the **estimate** that is furthest away, since 20 is further from 19.173 than 19 and 300 is further from 338 than 350.*

3) **Estimate** the following, making sure to show your working out.

e.g.

$$\begin{aligned} \text{a) } \frac{5742 \times 34.7}{875.42} &\approx \frac{6000 \times 30}{900} \\ &\approx \frac{180\,000}{900} \\ &\approx 200 \end{aligned}$$

$$\begin{aligned} \text{b) } (61.35 - 84.2)^2 &\approx (60 - 80)^2 \\ &\approx (-20)^2 \\ &\approx 400 \end{aligned}$$

$$\begin{aligned} \text{c) } \frac{15.3 - 7.81}{0.24} &\approx \frac{20 - 10}{0.2} \\ &\approx \frac{10}{0.2} \\ &\approx 10 \div \frac{1}{5} \\ &\approx 10 \times 5 \\ &\approx 50 \end{aligned}$$

$$\begin{aligned} \text{d) } \frac{9.6 + 16.51}{19.3 + 9.51^2} &\approx \frac{10 + 20}{20 + 10^2} \\ &\approx \frac{30}{20 + 100} \\ &\approx \frac{30}{120} \\ &\approx \frac{1}{4} \end{aligned}$$

$$\begin{aligned} \text{e) } \frac{11.049 - 4.21}{0.82 - 0.301} &\approx \frac{10 - 4}{0.8 - 0.3} \\ &\approx \frac{6}{0.5} \\ &\approx 6 \div \frac{1}{2} \\ &\approx 6 \times 2 \\ &\approx 12 \end{aligned}$$

$$\begin{aligned} \text{f) } 34^2 - \sqrt{139.82} &\approx 30^2 - \sqrt{144} \\ &\approx 900 - 12 \\ &\approx 888 \end{aligned}$$



Task B: Estimation in context

- 1) A science textbook uses comparisons with humans to show the sizes of different animals.
- a) Which of these is a good **estimate** for the width of the hand?

10 cm

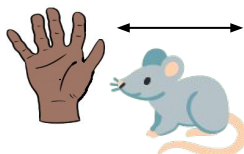
25 cm

5 cm



Use your answer to part a to **estimate** the size of the animals below.

b)



Length $\approx$ 15 cm

The rodent is approximately 1.5 times the width of a hand

c)



Length $\approx$ 3 cm

The spider is approximately a third of the width of a hand

- 2) a) What would be a sensible unit of mass to measure each of the items in the table?

baby 3.2 kg	elephant 4000 kg	Full suitcase 19 kg
Cargo ship container 2797 kg	newborn puppy 120 g	Packet of crisps 28 g

- b) Each of the values below matches one of the six items. Which is which?

4000	3.2	120	28	19	2797
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- c) An orange weighs approximately 120 g. How many oranges do you think would be the equivalent of one of each of the things in part a?

Baby: approximately 25 oranges

Elephant: approximately 32 000 oranges.

Full suitcase: approximately 160 oranges

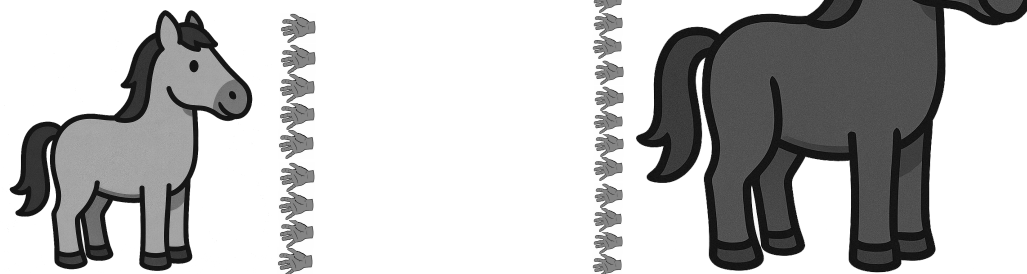
Cargo ship container: approximately 24 000 oranges

Newborn puppy: approximately 1 orange

Packet of crisps: approximately $\frac{1}{4}$ orange



3) Horses are measured using 'hands' as a unit, as shown below.



a) **Estimate** the height in centimetres of each of the horses.

*From left to right: 9 hands and 17 hands. Answers will vary depending on the **estimate** for the width of a hand: $9 \times \text{their estimate}$, $17 \times \text{their estimate}$. So, for example, if hand is **estimated** as 10 cm the answers will be 90 cm and 170 cm.*

b) The standard metric equivalent for a hand is 10.2 cm. Does this mean your **estimate** will be too big, too small or just right?

*If you used an **estimate** length of a hand to be less than 10.2 then your **estimate** will be too small, if you used a length greater than 10.2 cm then your **estimate** will be too big.*

c) How many hands high are you to the top of your head?

Answers will vary.

d) **Estimate** some other objects using hands as a unit.

Answers will vary.

